Latent Structure of Financial Development Indicators: Exploratory and Confirmatory Factor Analysis

1 Motivation

The principal component analysis in the main report reduced the 23 GFDD indicators to orthogonal components for prediction, but it did not test the *theoretical* claim embedded in the data design: that the indicators measure four latent constructs—**Depth, Access, Efficiency, Stability**. PCA maximises variance and assumes no measurement model; it cannot tell us whether the proposed four-dimension taxonomy actually holds. This addendum addresses that question directly with exploratory factor analysis (EFA), which lets the data suggest a latent structure, and confirmatory factor analysis (CFA), which formally tests whether the theoretical measurement model fits.

2 Factorability and Redundancy

Table 1: Near-duplicate indicator pairs ($|r| > 0.9$). These collinearities dominate the covariance structure and destabilise any measurement model that loads both members of a pair on the same factor.

Indicator A	Indicator B	r
di08 (Financial deposits / GDP)	oi02 (Bank deposits / GDP)	1.000
ei05 (ROA after tax)	ei09 (ROA before tax)	0.972
di01 (Private credit / GDP)	di02 (Bank assets / GDP)	0.971
di01 (Private credit / GDP)	di14 (Domestic credit / GDP)	0.968
ei06 (ROE after tax)	ei10 (ROE before tax)	0.952
di02 (Bank assets / GDP)	di14 (Domestic credit / GDP)	0.948
di05 (Liquid liabilities / GDP)	di08 (Financial deposits / GDP)	0.934
di05 (Liquid liabilities / GDP)	oi02 (Bank deposits / GDP)	0.934

The Kaiser–Meyer–Olkin measure of sampling adequacy is 0.7 (mediocre but above the 0.60 threshold), and Bartlett's test of sphericity rejects the identity-correlation null ($\chi^2 = 4001$, $df = 253$, $p < 0.001$), so the data are factorable. The dominant feature of the correlation matrix, however, is extreme redundancy: 8 indicator pairs correlate above 0.9 (Table 1), including a **perfect duplicate**—financial-system deposits (di08) and bank deposits (oi02) are identical ($r = 1.000$). Three credit measures (di01, di02, di14) are mutually correlated at ≈ 0.97, and the before-/after-tax profitability pairs at ≥ 0.95. Before the EFA, the perfect duplicate (oi02) and the before-tax profitability copies (ei09, ei10) are removed, leaving 20 indicators.

3 Exploratory Factor Analysis

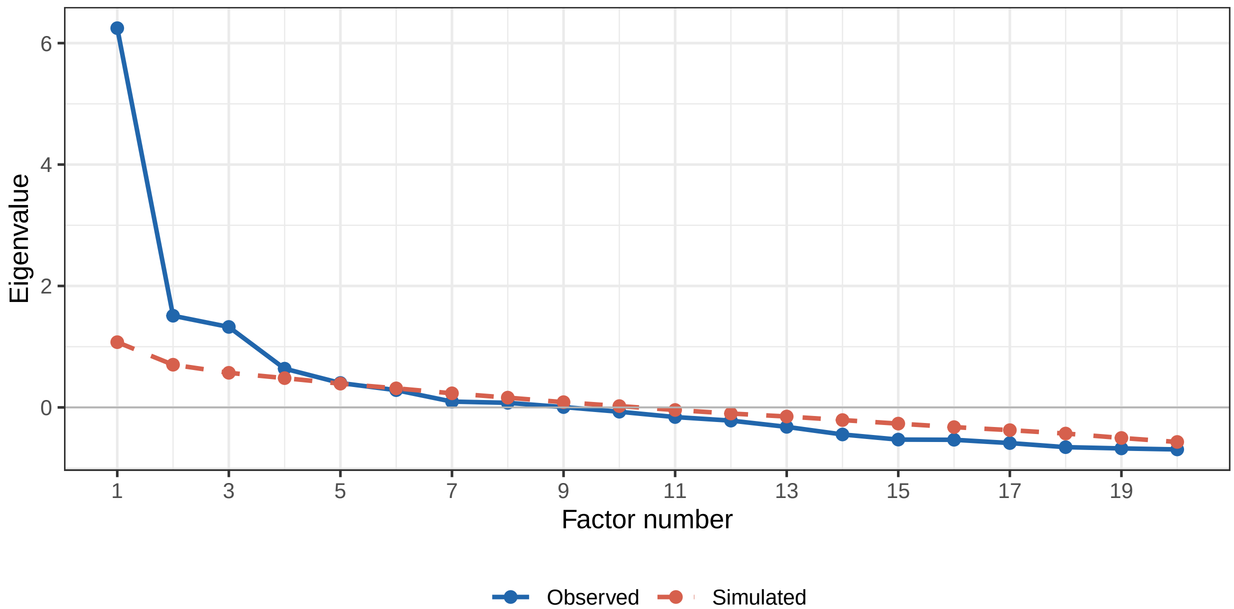

Figure 1: Parallel analysis. Solid line: eigenvalues from the 20-indicator factor solution; dashed line: 95th-percentile eigenvalues from random resampled data. Factors are retained where the observed eigenvalue exceeds the simulated benchmark.

Parallel analysis retains **4 factors** (the Velicer MAP criterion suggests 4); four factors are extracted to match the theoretical count. With oblimin (oblique) rotation—appropriate because financial dimensions are expected to correlate—the four factors explain 55% of the variance. The loading pattern (Figure 2) does **not** reproduce the theoretical Depth/Access/Efficiency/ Stability blocks:

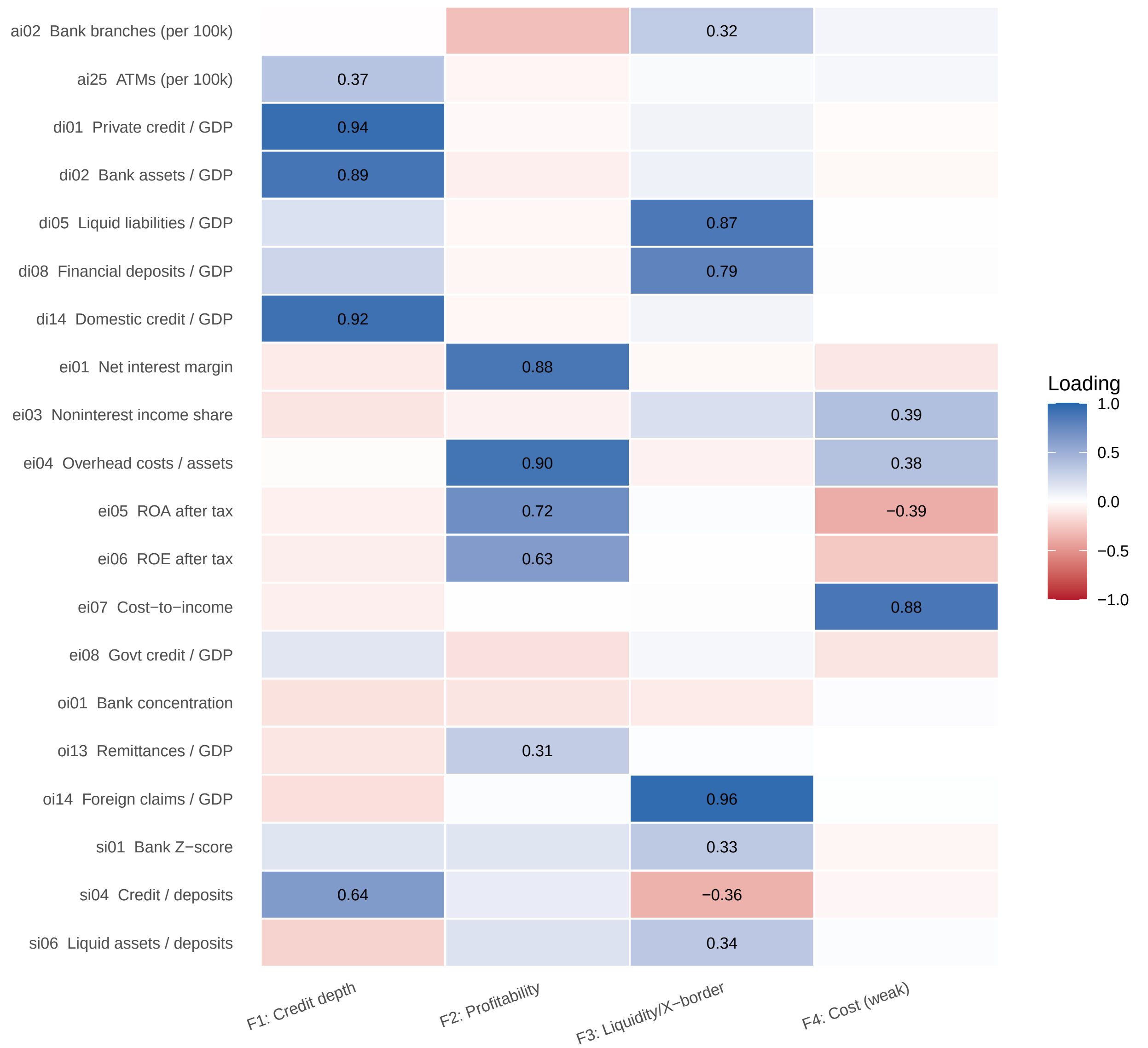

Figure 2: EFA loading heatmap (oblimin rotation, ML extraction). Cells show the loading of each indicator (rows) on each rotated factor (columns); values with $|\lambda| \geq 0.30$ are printed. The theoretical Depth block splits across two factors (Credit vs. Liquidity), Efficiency collapses to a Profitability factor, and the Access and Stability indicators load weakly everywhere.

- **F1 — Credit depth** loads private credit, bank assets and domestic credit (di01, di02, di14), plus the credit-to-deposits ratio (si04).
- **F2 — Profitability** loads net interest margin, overhead, and ROA/ROE (ei01, ei04, ei05, ei06)—the coherent core of the theoretical Efficiency block.
- **F3 — Liquidity / cross-border** loads liquid liabilities, deposits and foreign claims (di05, di08, oi14).
- **F4** is a weak doublet (cost-to-income, noninterest income) explaining only 7% of variance.

Crucially, the theoretical **Depth** dimension is *not* unidimensional—it splits into a credit factor (F1) and a liquidity factor (F3). The theoretical **Efficiency** dimension reduces to **Profitability**

(F2); its remaining indicators (noninterest income ei03, government credit ei08) are near-unique (communalities 0.2 and 0.1). The **Access** indicators (ai02, ai25, communalities ≤ 0.25) and the **Stability** indicators (si01, si06) fail to form common factors at all.

4 Confirmatory Factor Analysis

Model A — the literal theoretical four-factor model (Depth, Access, Efficiency, Stability) **fails to converge to an admissible solution** (the optimiser does not reach a proper solution): forcing the near-collinear depth indicators onto a single factor and the heterogeneous efficiency indicators (some negatively correlated) onto another produces Heywood cases. This non-convergence is itself the result—the proposed measurement model is empirically untenable as specified.

Model B — a respecified model based on the EFA evidence (Credit, Liquidity, Profitability, Access; the incoherent Stability factor dropped, the perfect duplicate removed) converges to a proper solution with the fit in Table 2.

Table 2: CFA fit statistics for the respecified four-factor model (Model B).

Fit index	Model B
χ^2 (df)	196.3 (49), p $<$ 0.001
CFI	0.905
TLI	0.872
RMSEA [90SRMR	0.080

Table 3: Model B standardised loadings. di05's residual variance is fixed at the Heywood boundary (0), so its loading is 1.00 by construction.

Factor	Indicator	Std. loading
Credit	di01 (Private credit / GDP)	0.99
	di02 (Bank assets / GDP)	0.98
	di14 (Domestic credit / GDP)	0.97
Liquidity	di05 (Liquid liabilities / GDP)	1.00
	di08 (Financial deposits / GDP)	0.93
	oi14 (Foreign claims / GDP)	0.84
Profit	ei01 (Net interest margin)	0.96
	ei04 (Overhead costs / assets)	0.85
	ei05 (ROA after tax)	0.77
	ei06 (ROE after tax)	0.65
Access	ai02 (Bank branches (per 100k))	0.80
	ai25 (ATMs (per 100k))	0.56

Model B fits only moderately: CFI = 0.91 and SRMR = 0.08 are acceptable, but RMSEA = 0.16 remains high, indicating residual misspecification that no simple-structure model fully removes given the indicators' redundancy. The Credit and Liquidity factors are extremely reliable ($\omega \geq 0.95$) but this reflects near-duplicate indicators rather than rich measurement; the Access factor is weak ($\omega = 0.64$, AVE $= 0.48 < 0.5$). The estimated factor correlations are substantial—Credit with Profitability -0.61 and Credit with Liquidity 0.64—confirming the constructs are related facets of overall financial development rather than independent dimensions.

Table 4: Composite reliability and average variance extracted (AVE) per factor.

Factor	ω (composite rel.)	AVE
Access	0.64	0.48
Credit	0.99	0.96
Liquidity	0.95	0.86
Profit	0.89	0.67

5 Conclusions

Factor analysis revises the picture from the main report in three ways. First, the **four theoretical dimensions are not recovered empirically**: the literal measurement model is inadmissible (CFA Model A), and the EFA instead yields a credit factor, a liquidity factor and a profitability factor, with access and stability indicators failing to cohere. Second, **"Depth" is at least two-dimensional**—credit provision and deposit/liquidity scale are statistically distinct constructs that happen to correlate. Third, the database carries **substantial measurement redundancy** (a perfect duplicate plus seven further pairs above $|r| = 0.9$); pruning to one indicator per redundant cluster would yield a cleaner, more interpretable instrument of roughly 12–14 indicators without information loss.

For the downstream analyses this implies that PCA scores (already orthogonal) remain the right input for LDA, but a parsimonious set of factor scores—Credit, Liquidity, Profitability—offers a more interpretable, theory-revising summary of cross-country financial structure than the four-fold taxonomy assumed at the outset.

6 Appendix: R Code

```
knitr::opts_chunk$set(
  echo    = FALSE,
  message = FALSE,
  warning = FALSE,
  fig.align = "center"
)
library(tidyverse)
library(readxl)
library(psych)
library(lavaan)
library(knitr)
library(kableExtra)

# The data file is an xlsx archive saved with a .csv extension.
tmp <- tempfile(fileext = ".xlsx")
invisible(file.copy("Proposal/20220909-global-financial-development-database.csv", tmp))
raw <- read_excel(tmp, sheet = "Data - August 2022")
vars23 <- c(
  "di01","di02","di05","di08","di14",          # Depth (5)
  "ai02","ai25",                                 # Access (2)
  "ei01","ei03","ei04","ei05","ei06",
  "ei07","ei08","ei09","ei10",                   # Efficiency (9)
  "si01","si04","si06",                          # Stability (3)
  "oi01","oi02","oi13","oi14"                    # Other (4)
)

df2019 <- raw |>
  filter(year == 2019) |>
  dplyr::select(iso3, country, region, income, all_of(vars23)) |>
  drop_na()

X <- scale(as.matrix(df2019[, vars23]))   # standardised 111 x 23

desc_lab <- c(
  di01="Private credit / GDP",        di02="Bank assets / GDP",
  di05="Liquid liabilities / GDP",    di08="Financial deposits / GDP",
  di14="Domestic credit / GDP",       ai02="Bank branches (per 100k)",
  ai25="ATMs (per 100k)",             ei01="Net interest margin",
  ei03="Noninterest income share",    ei04="Overhead costs / assets",
  ei05="ROA after tax",               ei06="ROE after tax",
  ei07="Cost-to-income",              ei08="Govt credit / GDP",
  ei09="ROA before tax",              ei10="ROE before tax",
  si01="Bank Z-score",                si04="Credit / deposits",
  si06="Liquid assets / deposits",    oi01="Bank concentration",
  oi02="Bank deposits / GDP",         oi13="Remittances / GDP",
```

```
  oi14="Foreign claims / GDP"
)
R    <- cor(X)
kmo  <- KMO(R)
bart <- cortest.bartlett(R, n = nrow(X))

# Near-duplicate indicator pairs (|r| > 0.9)
hi  <- which(abs(R) > 0.9 & upper.tri(R), arr.ind = TRUE)
dup_df <- data.frame(
  `Indicator A` = paste0(rownames(R)[hi[, 1]], " (", desc_lab[rownames(R)[hi[, 1]]], ")"),
  `Indicator B` = paste0(colnames(R)[hi[, 2]], " (", desc_lab[colnames(R)[hi[, 2]]], ")"),
  r = round(R[hi], 3),
  check.names = FALSE
)
dup_df <- dup_df[order(-dup_df$r), ]
# De-duplicate before EFA: drop the perfect duplicate oi02 (= di08) and the
# before-tax profitability copies ei09 (= ei05), ei10 (= ei06).
keep <- setdiff(vars23, c("oi02", "ei09", "ei10"))
Xd   <- X[, keep]

set.seed(1)
# fa.parallel() and VSS() print to the console; capture and discard that output.
invisible(capture.output({
  pa  <- fa.parallel(Xd, fa = "fa", fm = "ml", plot = FALSE)
  map <- VSS(Xd, n = 8, rotate = "oblimin", fm = "ml", plot = FALSE)
}))
n_map <- which.min(map$map)
efa <- fa(Xd, nfactors = 4, rotate = "oblimin", fm = "ml")

# Tidy loading matrix for the heatmap / table (label by factor role)
L <- unclass(loadings(efa))
fac_names <- c(ML2 = "F1: Credit depth",
               ML1 = "F2: Profitability",
               ML3 = "F3: Liquidity/X-border",
               ML4 = "F4: Cost (weak)")
colnames(L) <- fac_names[colnames(L)]
L <- L[, c("F1: Credit depth","F2: Profitability",
           "F3: Liquidity/X-border","F4: Cost (weak)")]

load_long <- as.data.frame(L) |>
  rownames_to_column("ind") |>
  pivot_longer(-ind, names_to = "Factor", values_to = "loading") |>
  mutate(label = desc_lab[ind],
         strong = abs(loading) >= 0.30)
dfc <- as.data.frame(X)

# Model A: literal theoretical four-factor congeneric model.
```

```r
modA <- '
  Depth      =~ di01 + di02 + di05 + di08 + di14
  Access     =~ ai02 + ai25
  Efficiency =~ ei01 + ei03 + ei04 + ei05 + ei06 + ei07 + ei08
  Stability  =~ si01 + si04 + si06
'
fitA      <- cfa(modA, data = dfc, std.lv = TRUE)
A_conv    <- lavInspect(fitA, "converged")

# Model B: respecified empirical model. Perfect duplicate oi02 removed; the
# incoherent Stability indicators (si01 vs si06: r = -0.07) dropped; di05's
# residual variance fixed to 0 (Heywood boundary from di05/di08 r = 0.93).
modB <- '
  Credit    =~ di01 + di02 + di14
  Liquidity =~ di05 + di08 + oi14
  Profit    =~ ei01 + ei04 + ei05 + ei06
  Access    =~ ai02 + ai25
  di05 ~~ 0*di05
'
fitB  <- cfa(modB, data = dfc, std.lv = TRUE)
fmB   <- fitMeasures(fitB, c("chisq","df","pvalue","cfi","tli",
                             "rmsea","rmsea.ci.lower","rmsea.ci.upper","srmr"))
ssB   <- standardizedSolution(fitB)

# Composite reliability (omega) and AVE per factor
ldB <- ssB[ssB$op == "=~", ]
rel_df <- ldB |>
  group_by(Factor = lhs) |>
  summarise(
    Omega = sum(est.std)^2 / (sum(est.std)^2 + sum(1 - est.std^2)),
    AVE   = mean(est.std^2),
    .groups = "drop"
  )
kable(dup_df,
      format    = "latex",
      booktabs = TRUE,
      row.names = FALSE,
      caption   = paste0("Near-duplicate indicator pairs ($|r| > 0.9$). ",
                        "These collinearities dominate the covariance structure ",
                        "and destabilise any measurement model that loads both ",
                        "members of a pair on the same factor."),
      linesep   = "") |>
  kable_styling(latex_options = c("hold_position","scale_down"), font_size = 8)
pa_df <- data.frame(
  Factor    = seq_along(pa$fa.values),
  Observed  = pa$fa.values,
  Simulated = pa$fa.sim
```

```
)
pa_df |>
  pivot_longer(-Factor, names_to = "Type", values_to = "Eigenvalue") |>
  ggplot(aes(Factor, Eigenvalue, colour = Type, linetype = Type)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.8) +
  geom_hline(yintercept = 0, colour = "grey70", linewidth = 0.3) +
  scale_colour_manual(values = c(Observed = "#2166ac", Simulated = "#d6604d")) +
  scale_linetype_manual(values = c(Observed = "solid", Simulated = "dashed")) +
  scale_x_continuous(breaks = seq(1, 20, 2)) +
  labs(x = "Factor number", y = "Eigenvalue") +
  theme_bw(base_size = 10) +
  theme(legend.position = "bottom", legend.title = element_blank())
ggplot(load_long, aes(Factor, fct_rev(paste0(ind, "  ", label)), fill = loading)) +
  geom_tile(colour = "white", linewidth = 0.4) +
  geom_text(data = filter(load_long, strong),
            aes(label = sprintf("%.2f", loading)), size = 2.6) +
  scale_fill_gradient2(low = "#b2182b", mid = "white", high = "#2166ac",
                       midpoint = 0, limits = c(-1, 1), name = "Loading") +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 9) +
  theme(axis.text.x = element_text(angle = 20, hjust = 1),
        panel.grid = element_blank())
fit_tab <- data.frame(
  Index = c("$\\chi^2$ (df)", "CFI", "TLI", "RMSEA [90% CI]", "SRMR"),
  Value = c(
    sprintf("%.1f (%d), p < 0.001", fmB["chisq"], fmB["df"]),
    sprintf("%.3f", fmB["cfi"]),
    sprintf("%.3f", fmB["tli"]),
    sprintf("%.3f [%.3f, %.3f]", fmB["rmsea"], fmB["rmsea.ci.lower"], fmB["rmsea.ci.upper"]),
    sprintf("%.3f", fmB["srmr"])
  )
)
kable(fit_tab, format = "latex", booktabs = TRUE, row.names = FALSE,
      escape = FALSE, col.names = c("Fit index", "Model B"),
      caption = "CFA fit statistics for the respecified four-factor model (Model B).",
      linesep = "") |>
  kable_styling(latex_options = "hold_position", font_size = 10)
load_tab <- ldB |>
  transmute(Factor = lhs,
            Indicator = paste0(rhs, " (", desc_lab[rhs], ")"),
            `Std. loading` = sprintf("%.2f", est.std))
kable(load_tab, format = "latex", booktabs = TRUE, row.names = FALSE,
      caption = "Model B standardised loadings. di05's residual variance is fixed at the Heywoo
      linesep = "") |>
  kable_styling(latex_options = "hold_position", font_size = 9) |>
  collapse_rows(columns = 1, latex_hline = "major")
```

```r
rel_print <- rel_df |>
  mutate(Omega = sprintf("%.2f", Omega), AVE = sprintf("%.2f", AVE))
kable(rel_print, format = "latex", booktabs = TRUE, row.names = FALSE,
      col.names = c("Factor", "$\\omega$ (composite rel.)", "AVE"),
      escape = FALSE,
      caption = "Composite reliability and average variance extracted (AVE) per factor.",
      linesep = "") |>
  kable_styling(latex_options = "hold_position", font_size = 10)
```